Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- **Subspace**
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- Introduction to Orthogonal Projections
  Reading • 40 min

Subspace

0:05 / 0:35 2x

A **subspace** U of a vector space V is a subset that is itself a vector space under the same operations as V. For U to be a subspace, it must satisfy the following conditions:

1. Contain the zero vector : $0 \in U$
2. Be closed under addition : For all $u, v \in U, u + v \in U$
3. Be closed under scalar multiplication : For all $\alpha \in R$ and $u \in U, \alpha \cdot u \in U$

Examples

- The trivial subspaces are {0} and V itself.
- The solution set of a homogeneous linear system $Ax = 0$ is a subspace of $\mathbb{R}^n$.
- A plane through the origin in $\mathbb{R}^3$ is a subspace.

Go to next item ✓ Completed

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